EE5239 Homework 3

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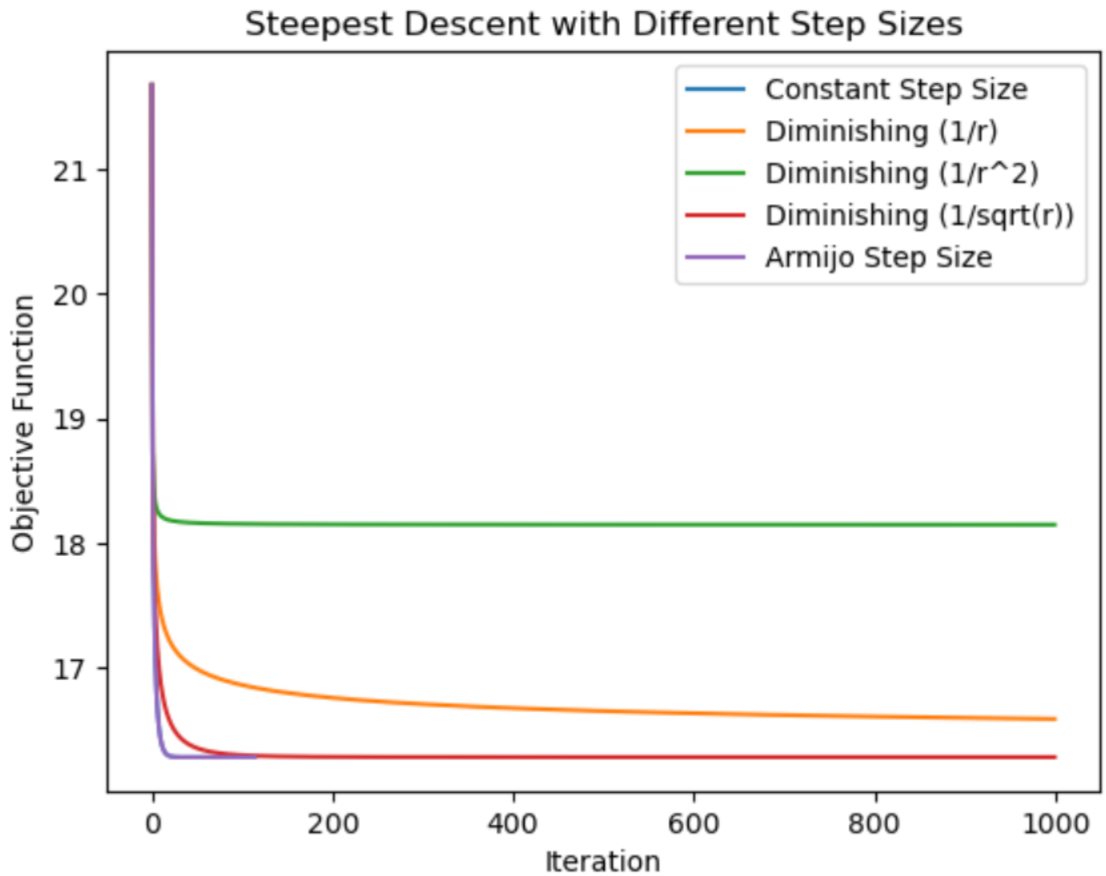
1 Problem 1

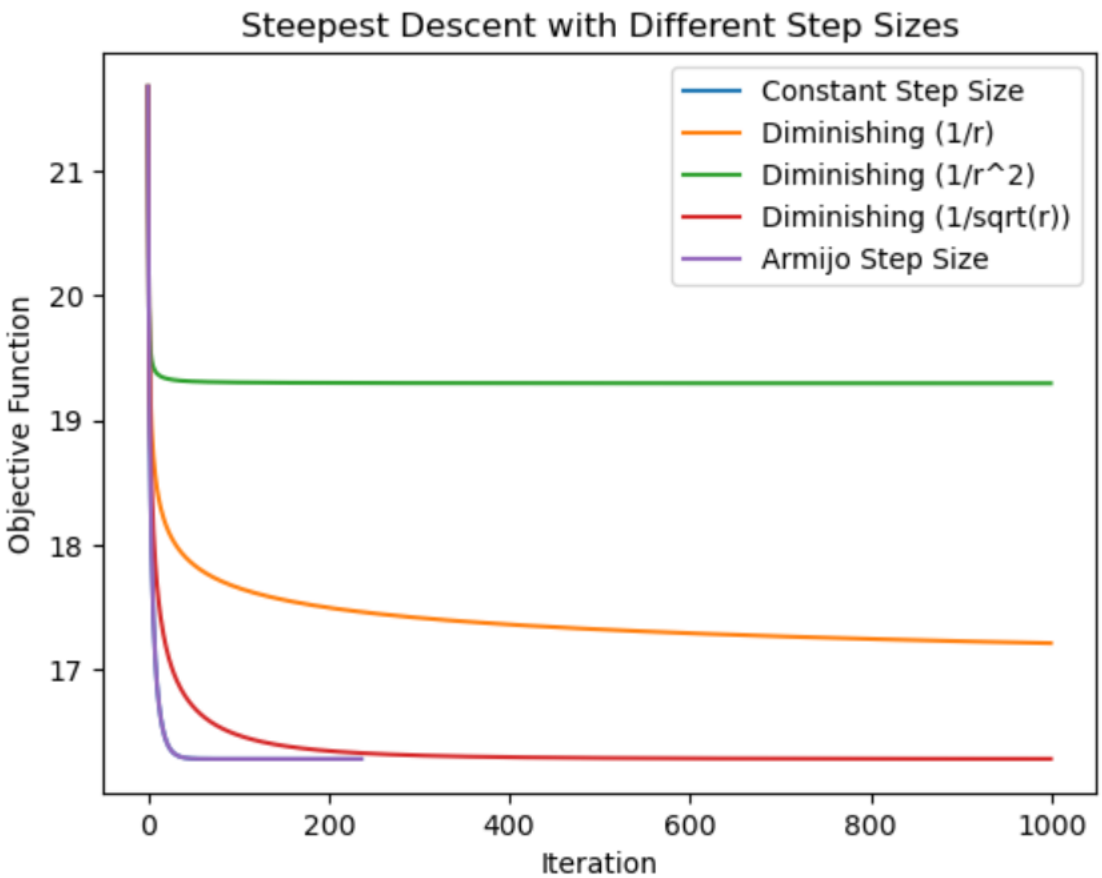
The problem I am solving here is to find the values of x that minimize $\|Ax - b\|^2$ where A is a randomly generated 50x10 matrix and b is a randomly generated vector of length 50. The algorithm that solves the problem is steepest descent, which subtracts the gradient times a step size variable from the previous iteration's result in each iteration. The step size variable for choice s is s, s/r, s/r^2 and s/sqrt(r) where r is the iteration number, as well as the armijo rule step sizes using alpha=s.

Stepsize choice 1: 0.01



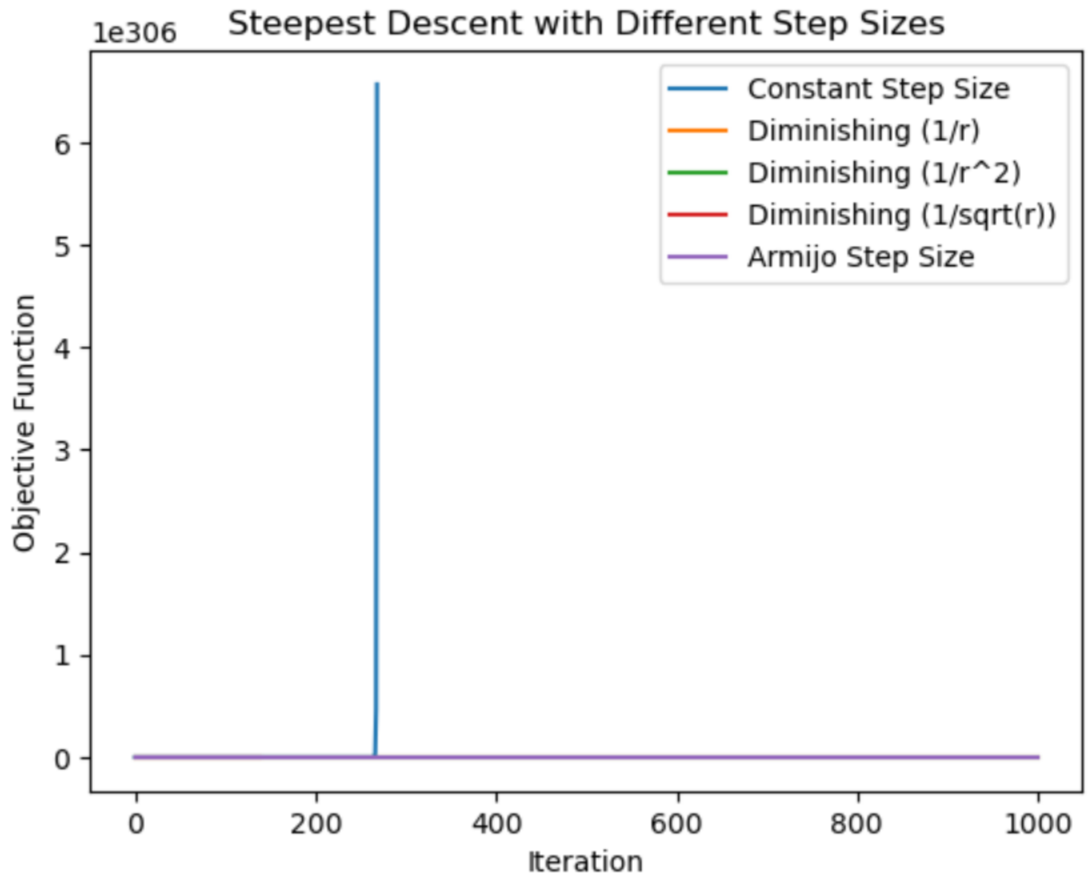
For this step size, the constant step size, armijo rule step size, and 1/sqrt(r) step size converge, while the others converge to a higher value, implying that the step size became too small before they reached the minimum.

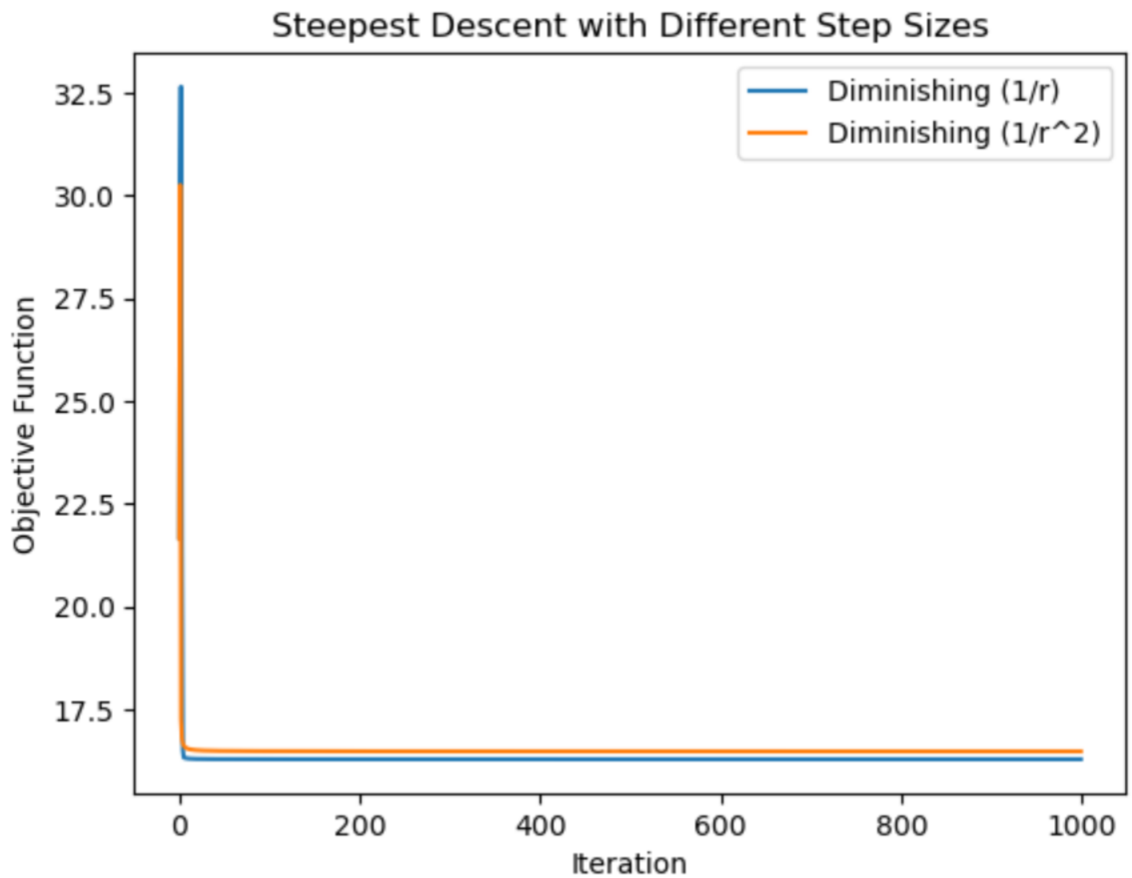
Stepsize choice 2: 0.005



For this step size, the constant step size, armijo rule step size, and 1/sqrt(r) step size converge, while the others converge to a higher value, implying that the step size became too small before they reached the minimum.

Stepsize choice 3: 0.05


Steepest Descent with Different Step Sizes
1e306
Objective Function
0
1
2
3
4
5
6
Iteration
0
200
400
600
800
1000
Constant Step Size
Diminishing (1/r)
Diminishing (1/r^2)
Diminishing (1/sqrt(r))
Armijo Step Size

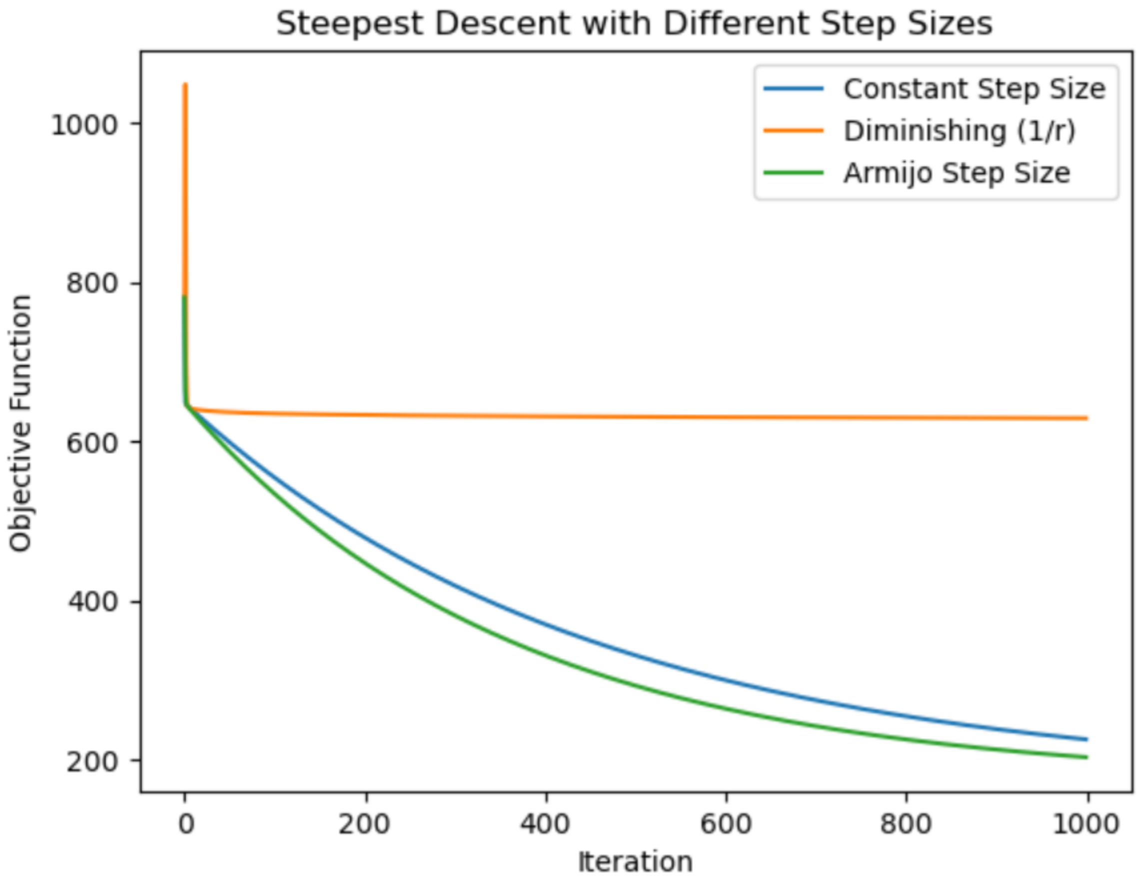


The second plot here includes only the step size choices that do not diverge.

For this step size, the constant step size, armijo rule step size, and 1/sqrt(r) step size diverge, while the others converge to the value to which the constant, armijo and 1/sqrt(r) step sizes converged earlier, implying that the step size was large enough this time that they no longer got stuck because the step size became too small before reaching the optimal solution.

2 Problem 2

The problem I am solving here is the same as that of part a, but with real data instead of randomly generated data. The matrix A is replaced by the 2 column matrix that represents the two predictor variables, symmetry and intensity, and the labels are replaced by the true digits. The data is filtered to only rows where the true digit is a one or a five and the model attempts to classify which is which. The methods used are the same as in part one except without the 1/sqrt(r) and $1/r^2$ step sizes.



This shows the convergence of each algorithm where the constant step size is 0.00008, the diminishing step size is 0.0002, and the armijo rule step size is 0.0001. The constant and armijo rule step sizes appear to converge at about the same rate while the diminishing step size does not converge to the optimal value, likely because its step size becomes too small as before.